EUROPEAN ALLIANCE FOR INDUSTRIAL DATA, EDGE AND CLOUD

THEMATIC ROADMAP

The Open Source Way to EU Digital Sovereignty & Competitiveness

A Strategic Roadmap for Action

Based on the July 2025 Report by the European Alliance for Industrial Data, Edge and Cloud

Who am I?

- Linux user since 1991
- Open Source advocate since 1998
- Open Source entrepreneur since 2000 (Nuxeo -> Abilian)
- Co-chair of CNLL, APELL
- Co-founder of EuroStack Initiative Foundation
- Member of the European Alliance for Industrial Data, Edge and Cloud, and co-author of the report

The Strategic Context: A Digital Crossroads

Europe's digital infrastructure is increasingly built on technologies developed and controlled outside our borders. What was considered by many a merely technical issue has now become a major strategic vulnerability.

This dependency creates significant risks for our:

- **Economic Autonomy**: Vulnerability to non-EU market dominance (€265 billion / year for cloud software and services from EU->US) .
- **Data Security**: Exposure to extraterritorial laws like the FISA and CLOUD Act, exposing personal information (cf. GDPR) and strategic / corporate information and secrets.
- **Strategic Independence**: A diminished ability to innovate according to our own values.


TECHNOLOGICAL DEPENDENCE ON AMERICAN SOFTWARE AND CLOUD SERVICES: AN ASSESSMENT OF THE ECONOMIC CONSEQUENCES IN EUROPE

Economic study

April 2025

ASTERèS

études, recherche & conseil économique

Defining Our Goal: Effective Digital Sovereignty

To be truly sovereign, we need more than just control over data. The report starts from a seimple and clear formula:

Digital Sovereignty = Data Sovereignty + Technological Autonomy

This means we must control the entire stack—the underlying hardware and software that manages our data. This technological independence is the key to effective sovereignty.

The Enabler: Why Open Source is the Cornerstone

Initially (1998-2010) consider mostly as a cheaper alternative, Open Source has become for the last 15 years a fundamental pillar of the technological autonomy Europe seeks.

- **Autonomy & Reversibility**
  It is the only real escape from vendor lock-in. Full control over the code guarantees the freedom to innovate independently.
- **Collaboration & Innovation**
  It allows us to pool resources to build cutting-edge technologies that no single entity could develop alone.
- **Transparency & Security**
  The ability to audit code allows for collective verification, vulnerability fixing, and ensures no hidden backdoors.

From Diagnosis to a Blueprint for Action

To build a credible plan, we needed a deep diagnosis. Our analysis of Europe's digital landscape revealed a fundamental disconnect between our ambition for sovereignty and the reality of our ecosystem.

This diagnosis identified **5 critical "gaps"**. From these gaps emerged **70 concrete proposals**, each designed to address a specific weakness and build a stronger whole.

Gap 1: Standards & Interoperability

The Challenge of "Open Washing"

The Problem: Our ecosystem is fragmented by "standards" that are not genuinely designed to promote interoperability and reversibility, promoted by non-EU players to maintain dominance. This creates vendor lock-in and stifles innovation.

The Consequence: A lack of genuine, enforceable interoperability that keeps Europe dependent and divided. Cf. the battle for EIFv2 (2009).

Gap 2: Financial Viability

The Peril of Instability

The Problem: Many critical European Open Source projects and business rely on sporadic funding, time-limited contracts and volunteer work. They lack resources for security audits, long-term maintenance, whole product development and professional support.

The Consequence: Key infrastructure components are unstable, uncompetitive, and at risk of being abandoned or acquired by non-EU interests.

Gap 3: Market Visibility & Adoption

The Fight for Recognition

The Problem: European solutions are often invisible, drowned out by the dominant marketing narratives of non-EU giants who sow doubt about the viability of Open Source.

The Consequence: Public and private sectors alike default to non-sovereign solutions, not because they are better, but because they are better known.

Gap 4: Talent & Skills Shortage

The Human Capital Deficit

The Problem: The lack of expertise in sovereign Open Source technologies slows down innovation and implementation.

The Consequence: A persistent skills gap makes us reliant on foreign expertise, directly undermining our goal of technological autonomy.

Gap 5: Governance & Influence

The Imbalance of Power

The Problem: Europe contributes a vast amount of code to global Open Source projects but exercises little control over their strategic direction, which is often decided in foundations based outside the EU.

The Consequence: Key technological dependencies remain, as the roadmaps of critical software are not aligned with European strategic interests.

The Blueprint: A 5-Pillar Strategy to Bridge the Gaps

Our 70 proposals are organized into a clear, actionable strategy built on five pillars. This is our roadmap for a sovereign digital future for Europe.

1. **Technological Development**
2. **Skills Development**
3. **Public Procurement**
4. **Growth and Investment**
5. **Governance**

(Origin, in large part: previous proposals and reports from APELL, CNLL, OSBA, paliamentary reports, European projects, etc.)

Pillar 1: Technological Development

Objective: To build a resilient technical foundation based on truly open standards and well-supported European projects.

Emblematic Actions:

1. **Create a European Open Source Sovereignty Fund (EOSSF)**
   To provide stable, long-term funding for the core infrastructure we all depend on.
2. **Define and Enforce TRULY Open Standards**
   Develop European-governed specifications to fight "open washing" and ensure genuine interoperability.
3. **Launch Large-Scale Demonstration Projects**
   Co-invest in pilot projects that prove the technical and economic viability of sovereign stacks in key sectors.

Pillar 2: Skills Development

Objective: To bridge the talent gap by fostering a skilled workforce proficient in European Open Source technologies.

Emblematic Actions:

1. **Establish European "Centres of Excellence"**
   Support universities in becoming hubs for advanced research, training, and innovation in sovereign open source technologies.
2. **Launch Pan-European Certifications**
   Create industry-recognized certifications for European Open Source skills, making them a gold standard for professionals and employers.
3. **Integrate Open Source into STEM Curricula**
   Embed open source principles and digital sovereignty concepts into education from secondary school onwards.

Pillar 3: Public Procurement

Objective: To leverage public spending as a powerful catalyst for the adoption of European Open Source solutions.

Emblematic Actions:

1. **Mandate "Public Money, Public Code, European Preference"**
   Adopt this powerful principle to ensure that taxpayer-funded software originates from, and benefits, the European ecosystem.
2. **Create a Public Directory of Sovereign Solutions**
   Build a trusted, curated repository of recommended European Open Source solutions to guide public procurement officers.
3. **Define Clear Criteria for "European Open Source"**
   Establish a robust definition to prevent misuse and ensure public funds support genuinely sovereign projects.

A Cornerstone Proposal: Defining "European Open Source"

To make policies like "European Preference" effective, we must first have a clear, robust definition to prevent "open washing".

Proposed Criteria for a "European Open Source" Solution:

- **Origin of Development**
  A significant portion of R&D and development is conducted in Europe.
- **Governance**
  The project's governance is based in Europe and aligned with EU values.
- **Community**
  A substantial part of the user and contributor base is European.
- **Legal Framework**
  Uses OSI-approved licenses compatible with EU law.
- **Data Handling**
  Prioritizes data processing and residency within the EU, in compliance with EU laws and directives (like GDPR, but non only).
- **Ecosystem Contribution**
  The maintaining organization actively contributes back to the European ecosystem.

From Definition to Action: A New Procurement Principle

Building on this definition, the report proposes a powerful new policy for public spending:

Public Money, Public Code,
Open Source First, <u>European Preference</u>

This extends the well-known "Public Money, Public Code" initiative. The **"European Preference"** clause ensures that taxpayer-funded software not only remains open, but also **originates from and benefits the European Open Source ecosystem**.

It transforms public procurement from a simple purchase into a strategic investment in our own digital future.

Pillar 4: Growth and Investment

Objective: To build a robust and sustainable funding ecosystem for the growth of European Open Source.

Emblematic Actions:

1. **Build a Strong "European Open Source" Brand**
   Develop a global brand that signifies quality, security, and alignment with European values, helping our solutions stand out.
2. **Launch a European OS Investment Platform (EOSIP)**
   Create a one-stop-shop portal to connect European Open Source projects with grants, venture capital, and other funding opportunities.
3. **Establish European Open Source Accelerators**
   Launch dedicated incubators to help promising projects scale up and compete globally.

Pillar 5: Governance

Objective: To ensure the long-term security, sustainability, and European control of critical Open Source projects.

Emblematic Actions:

1. **Form a European Open Source Advisory Council**
   Establish a high-level body of experts to guide funding, promote best practices, and ensure critical projects remain under European stewardship.
2. **Provide Support for Cyber Resilience Act (CRA) Compliance**
   Offer dedicated resources to help European projects achieve CRA certification, turning compliance into a mark of quality and trust.
3. **Promote Active Participation in Governance**
   Incentivize and support the active involvement of European experts in the decision-making bodies of key international foundations.

The Added Value: Transformative Impact on Key Sectors

Implementing this roadmap will unlock concrete benefits across our society:

- **Public Administration**
  Enhances security, transparency, and accountability while reducing dependency and lowering costs.
- **Manufacturing (Industry 4.0)**
  Enables predictive maintenance, improves interoperability on the factory floor, and fosters innovation.
- **Healthcare**
  Ensures patient data security, enables system interoperability, and accelerates medical research.
- **Energy**
  Supports a greener future by optimizing grids, integrating renewables, and reducing data center energy use.

A Vision for Europe: A Virtuous Cycle of Sovereignty

Implementing this roadmap will create a self-reinforcing cycle:

- **Sovereign Infrastructure** will give us control over our digital destiny.
- **A Thriving Ecosystem** will empower SMEs, create high-skilled jobs, and drive innovation.
- **Reduced Dependency** will make our economy more resilient and secure.
- **Technology Aligned with Our Values** will ensure that Europe's digital future is open, transparent, and human-centric.

Conclusion: A Call to Action

Achieving European digital sovereignty requires the collective support and action of our entire ecosystem—from policymakers and industry leaders to developers and educators. Additional initiatives include:

- The "Cloud Sovereignty Framework" from the Commission
- Various public procudement initiatives (EU and MS)
- The EU-STF (European Sovereign Tech Fund) initiative
- APELL's "Digital Declaration of Independence" (contact me)
- Etc.

Let's build our digital future, the Open Source way.

Get the full report: https://ec.europa.eu/newsroom/dae/redirection/document/117980